mechanisms to transfer the exception between threads. A sample implementation is shown here.

Listing 8.7. A parallel version of `std::for_each`

```
template<typename Iterator,typename Func>
void parallel_for_each(Iterator first,Iterator last,Func f)
{
    unsigned long const length=std::distance(first,last);

    if(!length)
        return;

    unsigned long const min_per_thread=25;
    unsigned long const max_threads=
        (length+min_per_thread-1)/min_per_thread;

    unsigned long const hardware_threads=
        std::thread::hardware_concurrency();

    unsigned long const num_threads=
        std::min(hardware_threads!=0?hardware_threads:2,max_threads);

    unsigned long const block_size=length/num_threads;

    std::vector<std::future<void> > futures(num_threads-1);      <-1
    std::vector<std::thread> threads(num_threads-1);
    join_threads joiner(threads);
```